

INTELLIGENT SYSTEMS COURSE PROJECT

FOG ORCHESTRATION • MQTT • DOCKER • IoT SMART NODES

Intelligent Fog Orchestration System

Distributed Fog Orchestration for IoT and Edge Workloads

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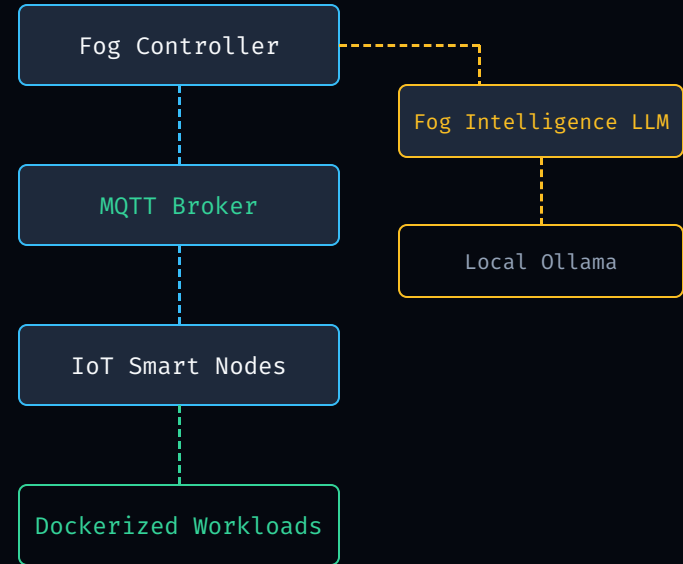
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System Scope

What Was Built

- Distributed fog orchestration platform, not only a dashboard.
- Fog Controller manages desired state, telemetry, events, and summaries.
- IoT Smart Nodes manage local Docker workloads.
- MQTT carries desired-state, telemetry, and event messages
- Dockerized workloads run across multiple node environments.



ARCHITECTURE OVERVIEW

SYSTEM ARCHITECTURE

- Fog Controller stores **desired state** and receives telemetry.
- MQTT Broker carries **desired-state**, **telemetry**, and **event messages**.
- IoT Smart Nodes manage local **Docker runtimes**.
- Workloads run as managed **Docker containers** on each node.
- Fog Intelligence LLM is **separate** from the MQTT orchestration path.

MQTT MESSAGE BUS

IoT Smart Node 1

ProMatch
FluidLab CFD

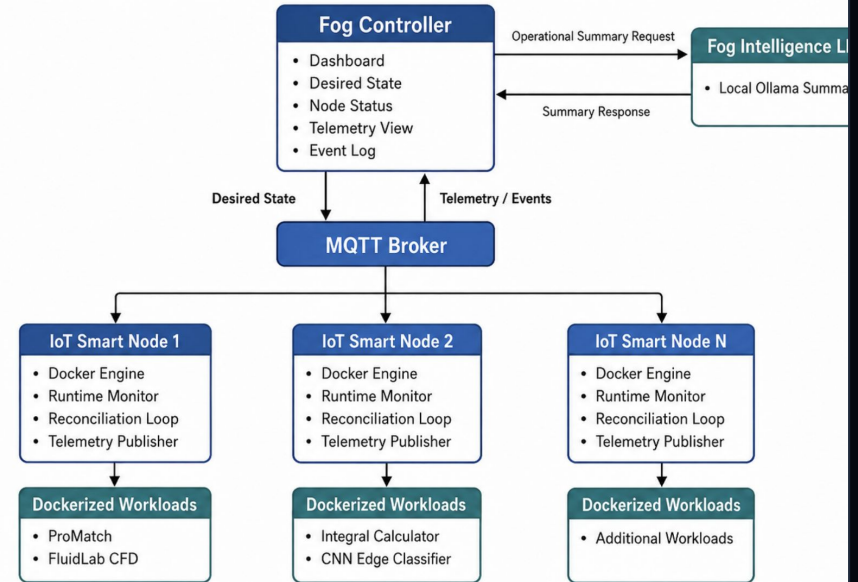
IoT Smart Node 2

Integral Calculator
CNN Edge Classifier

IoT Smart Node N

Additional
Workloads

Intelligent Fog Orchestration System – Architecture Overview



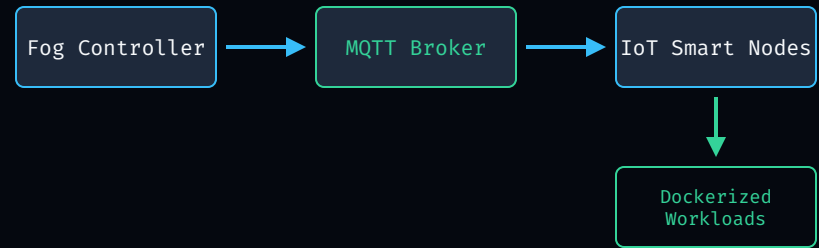
Scalable architecture: additional IoT Smart Nodes can be added with a new NODE_ID, MQTT configuration, and Docker access.

Runtime Architecture Paths

Runtime Paths

- Controller publishes desired-state updates through MQTT.
- IoT Smart Nodes publish telemetry and orchestration events back to the controller.
- Fog Intelligence LLM is called only by the controller.
- Workload deployment remains deterministic and is executed by the IoT Smart Node.

1. Main orchestration path



2. Operational summary path



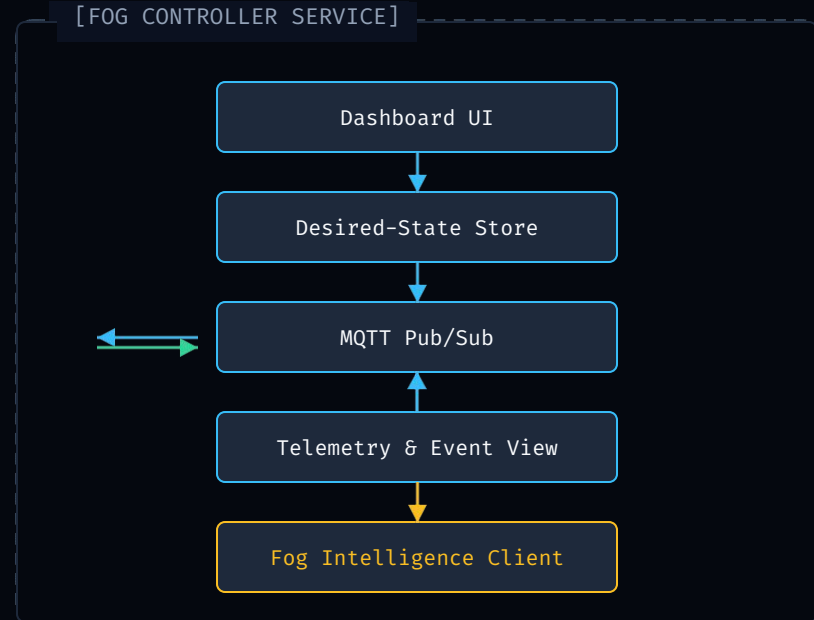
LLM support is outside the workload deployment path.



Fog Controller Architecture

Controller Responsibilities

- Stores desired workload state per node.
- Publishes desired-state updates through MQTT.
- Receives telemetry and orchestration events.
- Displays node status and workload state.
- Requests operational summaries from Fog Intelligence LLM.



Reconciliation is executed on IoT Smart Nodes, not inside the controller.

MQTT COMMUNICATION LAYER

MQTT MESSAGE BUS



Controller-Node Messaging

- MQTT Broker connects the Fog Controller and IoT Smart Nodes.
- Controller publishes desired-state updates.
- Nodes publish telemetry, workload status, and orchestration events.
- MQTT_HOST can point to local, LAN IP, or Tailscale IP.
- DEVICE_TOKEN protects controlled command paths.

```
mqtt_config.env
```

```
MQTT_HOST=host.docker.internal
```

```
MQTT_PORT=1883
```

```
// Connectivity Scope:
```

```
Local: Fog Controller ↔ MQTT Broker ↔ IoT Smart Nodes
```

```
Distributed: Fog Controller ↔ MQTT Broker at
```

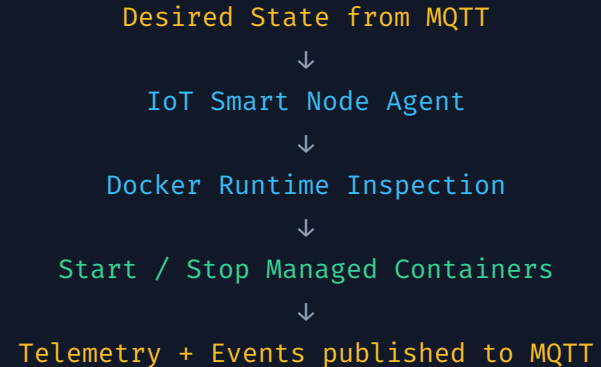
```
LAN/Tailscale IP ↔ IoT Smart Nodes
```



IoT Smart Node Architecture

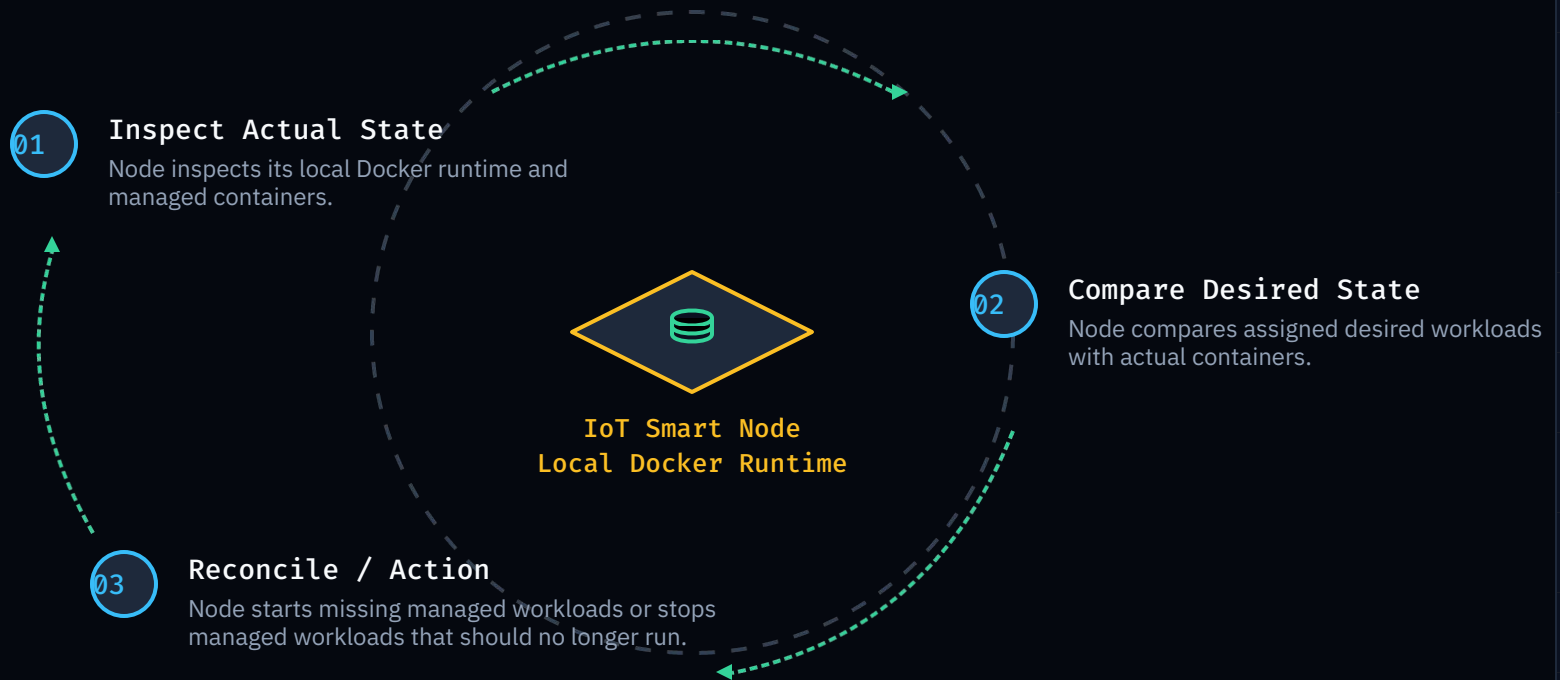
Node Responsibilities

- Runs with a fixed NODE_ID.
- Connects to the MQTT broker using MQTT_HOST and MQTT_PORT.
- Receives desired-state updates assigned to its node ID.
- Inspects the local Docker runtime through the Docker socket.
- Starts or stops managed workload containers.
- Publishes telemetry, workload status, and orchestration events.



** Docker socket access is required because the node manages containers on the host Docker runtime.*

Desired-State Reconciliation



Desired state is defined by the Fog Controller; reconciliation is executed locally by each IoT Smart Node.

|| Docker Workload Orchestration

Container Runtime Layer

- Workloads are packaged as Docker images.
- IoT Smart Node starts and stops managed containers locally.
- Docker socket access allows the node to control the host Docker runtime.
- Managed Docker labels identify orchestrated containers.
- Images are support linux/amd64 and linux/arm64.

└ ProMatch

└ FluidLab CFD

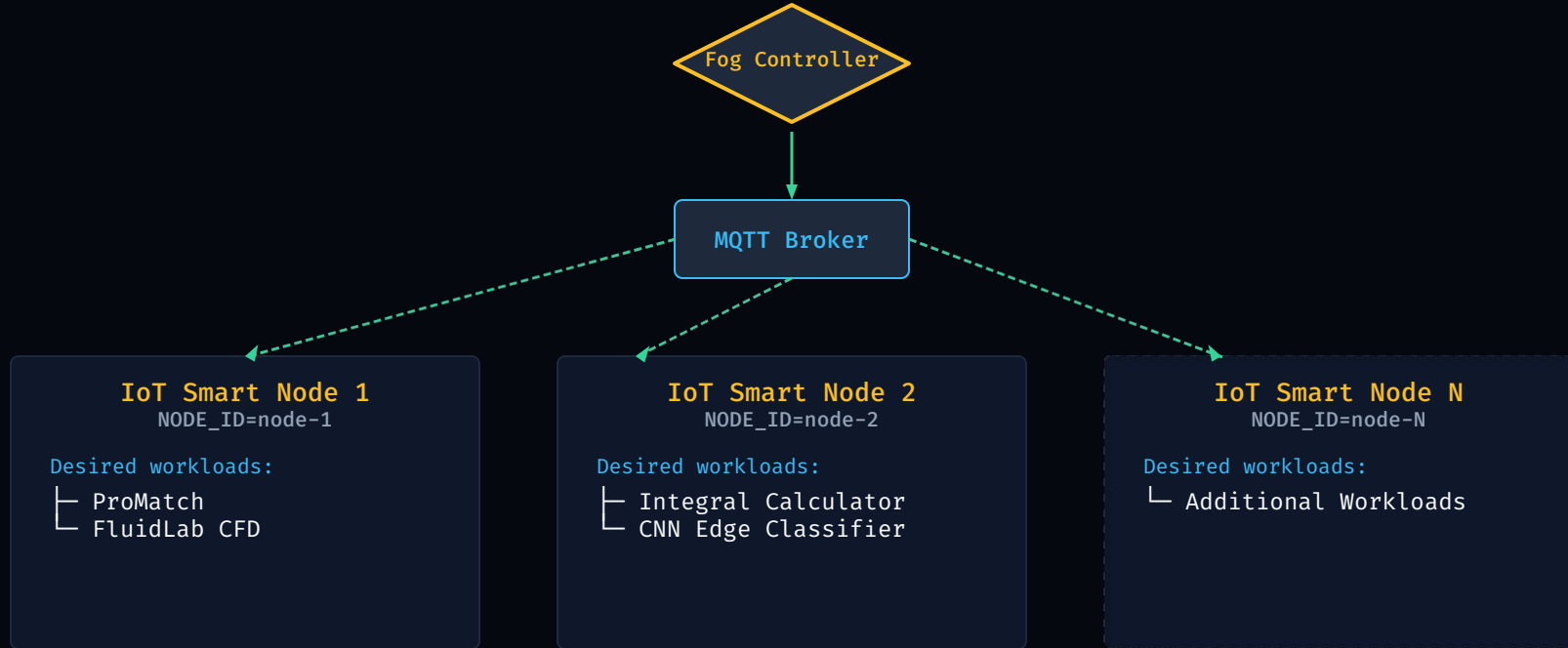
└ Integral Calculator

└ CNN Edge Classifier

** The IoT Smart Node manages workloads through the Docker runtime, not by knowing each workload's internal implementation.*

Workload Placement

Fog Controller publishes desired-state updates through MQTT.



The controller defines the desired placement; each node starts the assigned workloads on its own Docker runtime.



Heterogeneous Workload Layer

- One orchestration model manages different workload types.
- The IoT Smart Node controls containers through Docker, not each workload's internals stack.

ProMatch

Prolog reasoning / matchmaking

FluidLab CFD

Scientific simulation workload

Integral Calculator

Haskell symbolic computation and explanation support

CNN Edge Classifier

MobileNetV2 CNN/DCNN image inference

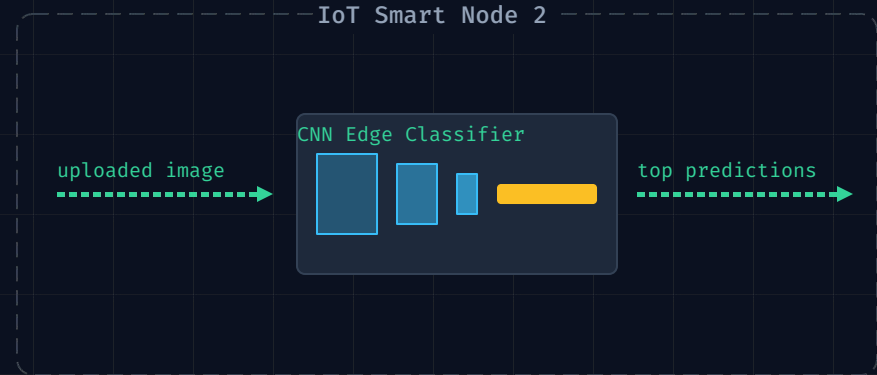
All workloads are managed through the same Dockerized workload interface.



CNN EDGE INFERENCE WORKLOAD

MobileNetV2 Inference Workload

- MobileNetV2 pretrained CNN model.
- Runs as a managed Docker workload on node-2.
- Provides a web UI and /predict API.
- Accepts uploaded images and returns top predictions, confidence scores, and inference time.
- Included to demonstrate CNN/DCNN edge inference within the fog orchestration platform.



The workload is orchestrated like the other services through Docker and desired-state assignment.

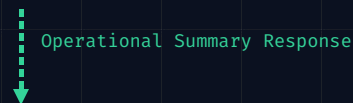
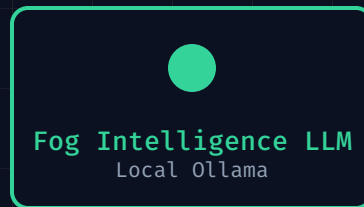


Fog Intelligence LLM

Operational Summary Support

- Fog Controller sends controller-level operational context to the LLM service.
- Fog Intelligence LLM connects to local Ollama through `OLLAMA_BASE_URL` and `OLLAMA_MODEL`.
- The service returns readable operational summaries for the dashboard.
- It does not deploy workloads and does not communicate directly with IoT Smart Nodes.
- Integral Calculator uses Cerebras separately for explanation generation.

LLM support is advisory only, workload deployment is still handled by desired-state reconciliation.



Operational Summary

- Node status
- Running workloads
- Resource usage
- Recent orchestration events



Security and Reliability

Implemented Controls

- DEVICE_TOKEN protects controlled command paths.
- Rate limiting and burst limiting reduce command abuse.
- Restricted configuration updates limit unsafe runtime changes.
- Managed Docker labels identify orchestrated containers.
- Environment variables keep secrets outside source code.
- Desired-state reconciliation restores missing managed workloads.

Security

- DEVICE_TOKEN
- Rate limiting
- Restricted config updates
- Environment-based secrets

Reliability

- Runtime monitoring
- Docker-managed labels
- Desired-state reconciliation
- Workload restart after mismatch

The IoT Smart Node requires Docker socket access and is treated as a trusted runtime component.



Evaluation and Results

The system was evaluated through runtime verification of the implemented fog orchestration flow.

2 IoT Smart Nodes

Connected through MQTT

4 Workloads

Deployed through Docker

Reconciliation

Verified after workload mismatch

Multi-Architecture

linux/amd64 and
linux/arm64 verified

Verified System Behavior

- Fog Controller publishes desired-state updates and receives telemetry/events.
- IoT Smart Nodes apply assigned workload state locally through Docker.
- node-1 runs ProMatch and FluidLab CFD.
- node-2 runs Integral Calculator and CNN Edge Classifier.
- Fog Intelligence LLM provides controller-level operational summaries.
- Docker Hub manifests verified linux/amd64 and linux/arm64 support.